Model 6 (v3): Identification and Estimation with Heterogeneous Covariates

This document develops the identification theory for the structural model with observable heterogeneity (Model 6, `M6_heterogeneity_information.tex`) when the econometrician does not observe—or only partially observes—the public covariates $\mathbf{x}$ that both banks use.

Model 5 (`M5_convert4intostructural.tex`) establishes nonparametric identification when there is no observable heterogeneity: bank 2 plays a single rate, and three data objects $(r_2^*, \sigma^*(\cdot), D_i)$ suffice to recover (α, λ, F). Model 6 introduces public covariates $\mathbf{x}$ observed by both banks, which generates heterogeneity in bank 2's rate across borrowers. The central question is: <u>what can the econometrician recover when $\mathbf{x}$ is partially or fully unobserved?</u>

Section 1 recaps the model. Section 2 establishes the sufficient statistic result. Section 3 characterizes precisely what is and is not identified nonparametrically. Section 4 provides formal identification results under parametric assumptions, stating exact conditions. Section 5 derives the likelihood. Section 6 describes the Monte Carlo validation.

1 Model Recap

Two banks compete for a borrower. Bank 1 (informed/incumbent) observes both public covariates $\mathbf{x}$ and private covariates $\mathbf{x}'$, and therefore knows the borrower's default probability $h = h(\mathbf{x}, \mathbf{x}')$. Bank 2 (uninformed/entrant) observes only $\mathbf{x}$ and faces the conditional distribution $F(h \mid \mathbf{x})$. Demand is logit:

$$s(r_1, r_2) = \frac{1}{1 + \exp\big(\alpha(r_1 - r_2) - \lambda\big)} \tag{1}$$

In equilibrium:

$$\sigma^*(h; \mathbf{x}) = \frac{1}{1-h} + \frac{1}{\alpha\big(1 - s(\sigma^*(h;\mathbf{x}), r_2^*(\mathbf{x}))\big)} \tag{2}$$

$$r_2^*(\mathbf{x}) = \frac{1}{1-\tilde{h}(\mathbf{x})} + \frac{1}{\alpha} \cdot \frac{\int (1-s)(1-h)\, dF(\cdot|\mathbf{x})}{\int s\,(1-s)(1-h)\, dF(\cdot|\mathbf{x})} \tag{3}$$

The structural parameters are $\theta = (\alpha, \lambda, F(\cdot|\mathbf{x}))$.

Data. For each borrower $i = 1, \ldots, N$, the econometrician observes:

Variable	Description	Values
r_i^{win}	Interest rate charged by the chosen bank	$\mathbb{R}_+$
W_i	Identity of the chosen bank	$\{1, 2\}$
D_i	Default indicator	$\{0, 1\}$
$\mathbf{z}_i$	Observed public covariates (possibly empty)	$\mathbb{R}^{d_z}$

We partition $\mathbf{x} = (\mathbf{z}, \mathbf{w})$, where $\mathbf{z}$ is observed by the econometrician and $\mathbf{w}$ is not. The analysis covers all cases from $\mathbf{z} = \mathbf{x}$ (full observability, nonparametric identification as in M5) to $\mathbf{z} = \emptyset$ (no covariates, most demanding).

Key data asymmetry:

- When $W_i = 2$: we observe $r_i^{\text{win}} = r_2^*(\mathbf{x}_i)$, directly revealing the entrant's rate for that borrower.
- When $W_i = 1$: we observe $r_i^{\text{win}} = \sigma^*(h_i; r_2^*(\mathbf{x}_i))$, which depends on two latent variables: the type h_i and the entrant rate $r_2^*(\mathbf{x}_i)$.

2 The Sufficient Statistic Result

Proposition 1 (Sufficient statistic)**.** *In the stage-2 equilibrium, the public covariates* $\mathbf{x}$ *affect bank 1's pricing rule* σ^* *and the market shares* s *only through the entrant's rate* $r_2^*(\mathbf{x})$*. Formally, for any two covariate vectors* $\mathbf{x}_a, \mathbf{x}_b \in \mathcal{X}$ *with* $r_2^*(\mathbf{x}_a) = r_2^*(\mathbf{x}_b) = r_2$*:*

$$\sigma^*(h; \mathbf{x}_a) = \sigma^*(h; \mathbf{x}_b) = \sigma^*(h; r_2), \qquad s(\sigma^*(h; \mathbf{x}_a), r_2) = s(\sigma^*(h; \mathbf{x}_b), r_2) \tag{4}$$

for all h*.*

Proof. Immediate from the bank 1 FOC (2): the right side depends on $\mathbf{x}$ only through $r_2^*(\mathbf{x})$. Since bank 1's best response is the unique solution to this fixed-point equation, and the equation is identical for all $\mathbf{x}$ with the same r_2^*, the solution $\sigma^*(h; \mathbf{x})$ is the same. □

Remark 2 (What r_2^* does not capture)*.* The sufficient statistic result applies to bank 1's strategy and market shares, but <u>not</u> to the full equilibrium. Bank 2's FOC (3) depends on $F(h|\mathbf{x})$, which is not a function of r_2^* alone—two different $\mathbf{x}$ values can lead to different $F(h|\mathbf{x})$ but the same optimal r_2^*.

Definition 3 (Type-recovery function)**.** For fixed r_2, define $\tau(r_1; r_2)$ as the inverse of $\sigma^*(\cdot\,; r_2)$:

$$\tau\big(\sigma^*(h; r_2);\, r_2\big) = h \qquad \text{for all } h \tag{5}$$

Since $\sigma^*(\cdot; r_2)$ is strictly increasing in h, $\tau(\cdot; r_2)$ is well-defined and strictly increasing in r_1. The Jacobian is:

$$\tau'(r_1; r_2) = \frac{1}{\sigma_h^*\big(\tau(r_1; r_2);\, r_2\big)} > 0 \tag{6}$$

3 What Is and Is Not Identified Nonparametrically

We now characterize precisely what can be recovered from the data without parametric assumptions, and where nonparametric identification breaks down. This makes explicit what the parametric assumptions in Section 4 are doing.

3.1 Benchmark: Full Observability ($\mathbf{z} = \mathbf{x}$)

When the econometrician observes all public covariates, conditioning on $\mathbf{x}$ reduces the model to M5 within each $\mathbf{x}$-stratum. The M5 identification result applies market-by-market:

1. $r_2^*(\mathbf{x})$ is identified as the common winning rate among bank-2 winners with covariates $\mathbf{x}$.
2. $\sigma^*(\cdot; \mathbf{x})$ is identified via type recovery: $E[D_i \mid r_i^{\text{win}} = r, W_i = 1, \mathbf{x}_i = \mathbf{x}] = \tau(r; r_2^*(\mathbf{x}))$.
3. (α, λ) are identified from the linear restriction $\ln(\alpha\, m - 1) = \lambda - \alpha\Delta$ within any single $\mathbf{x}$-stratum. Multiple strata provide overidentification.
4. $F(h|\mathbf{x})$ is identified by inverting the selection-weighted density of types among bank-1 winners.

This is the nonparametric ideal. We now analyze what survives when $\mathbf{x}$ is partially or fully unobserved.

3.2 What IS Identified Without x

Even without observing $\mathbf{x}$, several objects are nonparametrically identified from the data. Understanding these is important because they constrain any parametric model.

Proposition 4 (Identified objects without covariates)**.** *From the joint distribution of (r_i^{win}, W_i, D_i) alone, the following are nonparametrically identified:*

(a) The distribution of $r_2^*(\mathbf{x})$ among bank-2 winners (the selected distribution).

(b) The conditional default rate $E[D_i \mid r_i^{win}, W_i = 2]$ as a function of the winning rate.

(c) The distribution of $\sigma^*(h_i; r_2^*(\mathbf{x}_i))$ among bank-1 winners.

(d) The conditional default rate $E[D_i \mid r_i^{win}, W_i = 1]$ as a function of the winning rate. However, this does <u>not</u> recover the borrower's type h. The reason is that bank 1's rate depends on both the borrower's type and the competitive environment: $r_1 = \sigma^*(h; r_2^*(\mathbf{x}))$. Two borrowers with the <u>same</u> type h but different covariates $\mathbf{x}$ face different entrant rates $r_2^*(\mathbf{x})$, so bank 1 charges them different rates r_1. Conversely, the same r_1 can arise from different $(h, \mathbf{x})$ combinations: a safe borrower whose covariates $\mathbf{x}$ imply a competitive entrant (low r_2^*), or a riskier borrower whose covariates imply a less competitive entrant (high r_2^*). Since the econometrician does not observe $\mathbf{x}$, she cannot determine which competitive environment generated the observed r_1, and therefore cannot back out h. The conditional default rate $E[D_i \mid r_1, W_i = 1]$ is thus a weighted average over the different types that produce rate r_1, not the type itself.

(e) The aggregate switching probability $\Pr(W_i = 2)$.

Proof. These are all features of the observable joint distribution $(r_i^{\text{win}}, W_i, D_i)$ and are identified directly from the data without any model structure. □

Let us examine each and its informational content.

(a) Distribution of r_2^* among switchers. Among bank-2 winners, $r_i^{\text{win}} = r_2^*(\mathbf{x}_i)$, so the empirical distribution of winning rates directly reveals the selected distribution. However, this is not the population distribution of r_2^*: the density observed is

$$g_{r_2|W=2}(r_2) = \frac{g(r_2) \cdot \Pr(W = 2 \mid r_2)}{\Pr(W = 2)} \tag{7}$$

where $g(r_2)$ is the population density and $\Pr(W = 2 \mid r_2) = \int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2)$ is the aggregate switching probability conditional on r_2. Lower r_2 (more competitive entrant) means more switching, so the observed distribution is shifted downward relative to the population. Recovering $g(r_2)$ from $g_{r_2|W=2}(r_2)$ requires knowing $\Pr(W = 2|r_2)$, which depends on (α, λ, F).

(b) Default rates among bank-2 winners. For each value of the winning rate r_2:

$$E[D_i \mid r_i^{\text{win}} = r_2, W_i = 2] = \frac{\int h\,(1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2)}{\int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2)} \tag{8}$$

This is the adverse-selection-weighted mean type: borrowers who switch to bank 2 are disproportionately those with high h (risky types), because they face a higher $\sigma^*(h)$ that pushes them away from bank 1. Formally, $1 - s(\sigma^*(h), r_2)$ is increasing in h (since σ^* is increasing and $s_1 < 0$), so $E[D|r_2, W{=}2] > E[h|r_2]$. This object is directly estimable by regressing D_i on r_i^{win} among bank-2 winners.

(d) The mixture problem for bank-1 winners. Among bank-1 winners, the winning rate $r_1 = \sigma^*(h_i; r_2^*(\mathbf{x}_i))$ confounds two latent variables: h_i and $r_2^*(\mathbf{x}_i)$. The same r_1 can arise from different (h, r_2) combinations:

- Low h (good type) + low r_2 (competitive entrant): bank 1 offers a low rate against tough competition.
- Higher h (riskier) + higher r_2 (less competitive entrant): bank 1 charges more due to both higher cost and more market power.

The conditional default rate is a mixture:

$$E[D_i \mid r_i^{\text{win}} = r_1, W_i = 1] = \frac{\int_{\mathcal{R}_2(r_1)} \tau(r_1; r_2) \cdot s(r_1, r_2) \cdot f(\tau(r_1; r_2)|r_2) \cdot \tau'(r_1; r_2) \cdot g(r_2)\, dr_2}{\int_{\mathcal{R}_2(r_1)} s(r_1, r_2) \cdot f(\tau(r_1; r_2)|r_2) \cdot \tau'(r_1; r_2) \cdot g(r_2)\, dr_2} \tag{9}$$

This cannot be inverted to recover h_i for any individual borrower without additional structure.

3.3 What Fails Without x: The Benchmark Argument

Without observing $\mathbf{x}$, the four-step nonparametric identification argument from the benchmark case (Section 3.1) breaks down:

(i) **Step 1 (Recovery of r_2^*):** Bank 2's rate is no longer a single constant. It varies across borrowers through $\mathbf{x}$, generating a distribution rather than a point mass among bank-2 winners.

(ii) **Step 2 (Type recovery):** The key identity $E[D_i \mid r_1, W_i = 1] = h_i$ fails. The source of the problem is heterogeneity in the competitive environment through $\mathbf{x}$. Borrowers with covariates $\mathbf{x}$ that make the entrant very competitive (low $r_2^*(\mathbf{x})$) face a very different pricing environment from borrowers whose covariates make the entrant less competitive (high $r_2^*(\mathbf{x})$). Bank 1's rate $r_1 = \sigma^*(h; r_2^*(\mathbf{x}))$ depends on both the type and the competitive environment, and σ^* is increasing in both arguments. So the same r_1 can be generated by a low-risk borrower in a competitive market (low h, low r_2^*) or a higher-risk borrower in a less competitive market (high h, high r_2^*). Without observing $\mathbf{x}$, the econometrician cannot determine which competitive environment generated the observed rate, so the conditional expectation $E[D_i \mid r_1, W_i = 1]$ averages over these different types rather than recovering a single h.

(iii) **Step 3 (Identifying restriction):** The linear restriction $\ln(\alpha m - 1) = \lambda - \alpha\Delta$ requires knowing both $m(h) = \sigma^*(h) - 1/(1-h)$ and $\Delta(h) = \sigma^*(h) - r_2^*$ for individual borrowers. Without type recovery, neither m nor Δ can be computed.

(iv) **Step 4 (Distribution recovery):** Recovering $F(h)$ from the selection-weighted density $f_{h|W=1}(h)$ requires knowing $s(h)$, which requires (α, λ) from Step 3.

However, showing that a particular identification strategy fails does not establish that the model is unidentified—perhaps a different argument could work. We now prove that no such argument exists.

3.4 Nonparametric Non-Identification

Proposition 5 (Nonparametric non-identification)**.** *Without parametric restrictions on $F(h|\mathbf{x})$ and the distribution of $\mathbf{x}$, the structural parameters (α, λ) are not identified. Specifically: for any parameter vector $(\alpha_0, \lambda_0, F_0, G_0)$ generating observable distribution $\mathcal{D}$, and for any (α_1, λ_1) in a neighborhood of (α_0, λ_0), there exist nonparametric distributions F_1 and G_1 such that $(\alpha_1, \lambda_1, F_1, G_1)$ also generates $\mathcal{D}$.*

Proof. We construct an explicit observationally equivalent model. Fix $(\alpha_1, \lambda_1) \neq (\alpha_0, \lambda_0)$ with $\alpha_1 > 0$.

Step 1: New equilibrium. Under (α_1, λ_1), bank 1's FOC defines a new equilibrium strategy $\sigma_1^*(h; r_2)$ and market shares $s_1(r_1, r_2)$ for each r_2. These are uniquely determined (for given F and r_2, the bank-1 FOC has a unique solution under standard regularity).

Step 2: Matching bank-2 winner defaults. The data identify $\delta(r_2) \equiv E[D_i \mid r_2, W_i = 2]$ at each r_2. Under the alternative model:

$$\delta(r_2) = \frac{\int h\,(1 - s_1(\sigma_1^*(h; r_2), r_2))\, dF_1(h|r_2)}{\int (1 - s_1(\sigma_1^*(h; r_2), r_2))\, dF_1(h|r_2)}$$

This is a single moment restriction on the nonparametric distribution $F_1(\cdot|r_2)$ at each r_2. Since $F_1(\cdot|r_2)$ is a probability distribution on $[\underline{h}, \bar{h}]$—an infinite-dimensional object—a single moment does not pin it down. We can choose $F_1(\cdot|r_2)$ to satisfy $\delta(r_2)$ at each r_2, with many degrees of freedom remaining.

Step 3: Matching the r_2 distribution. Given $(\alpha_1, \lambda_1, F_1)$, the selection probability $\Pr_1(W = 2|r_2)$ is determined. Set:

$$g_1(r_2) \propto \frac{g_{r_2|W=2}(r_2)}{\Pr_1(W=2|r_2)}$$

where $g_{r_2|W=2}$ is the observed density of r_2 among bank-2 winners. This ensures the alternative model matches the observed distribution of entrant rates among switchers.

Step 4: Matching bank-1 winner density. The density of (r_1, D) among bank-1 winners is a mixture:

$$p(r_1, d) = \int f_1(\tau_1(r_1; r_2) \mid r_2)\, \tau_1'(r_1; r_2)\, s_1(r_1, r_2)\, h^d (1-h)^{1-d}\, g_1(r_2)\, dr_2$$

Here $\tau_1(\cdot; r_2)$ is the type-recovery function under the alternative equilibrium. At each r_2, the integrand involves $f_1(\cdot|r_2)$, which has the remaining degrees of freedom from Step 2. The system of equations (one per r_1 value) constrains the marginal contributions at each r_2 through a Fredholm integral equation of the first kind. Since $f_1(\cdot|r_2)$ is unrestricted (beyond the single moment matched in Step 2), this system generically admits solutions: the operator maps an infinite-dimensional input to a one-dimensional output at each r_1.

Conclusion. For any (α_1, λ_1) near the truth, we can find nonparametric (F_1, G_1) matching all observable moments:

- The distribution of r_2 among bank-2 winners (Step 3),
- The conditional default rates among bank-2 winners (Step 2),
- The joint density of (r_1, D) among bank-1 winners (Step 4),
- The aggregate switching rate (absorbed by the normalization of G_1 in Step 3).

The fundamental reason is a <u>dimension mismatch</u>: the data provide functional restrictions (a continuum indexed by r_1 or r_2), but the nonparametric unknowns $F_1(h|r_2)$ constitute a family of distributions—one per r_2—which is a higher-dimensional object. Two scalar parameters (α, λ) cannot be pinned down when the nuisance functions absorb all identifying variation. □

Remark 6 (Intuition). The non-identification has a simple economic interpretation. Suppose the true α (price sensitivity) is moderate. An alternative model with higher α_1 would predict stronger adverse selection (riskier borrowers switch more), changing the predicted default rates. But this can be offset by choosing $F_1(h|r_2)$ with a different shape—for instance, a type distribution with less dispersion, so that even with stronger selection, the default rates among switchers match the data. With nonparametric F, there is always enough flexibility to absorb any change in (α, λ).

3.5 The Role of Default Data in Narrowing Identification

Default outcomes D_i provide additional identifying power beyond rates and winner identity. Two mechanisms:

1. **Among bank-2 winners:** The function $E[D_i \mid r_i^{\text{win}} = r_2, W_i = 2]$ is identified. Since this is a weighted average of types who switch at entrant rate r_2, it traces out how the adverse-selection-weighted mean type varies with r_2. A structural model must match this entire curve, not just its level—this is a functional restriction, not a scalar one.

2. **Among bank-1 winners:** Although $E[D_i \mid r_1, W_i = 1]$ is a mixture (not a clean type recovery), its shape is informative. For instance:

   - If the default rate rises steeply with r_1, this is consistent with a narrow range of r_2^* (the mixture is concentrated, close to a clean type recovery).
   - If the default rate is flat or non-monotone, this indicates substantial variation in r_2^* (the mixture blurs the h–r_1 mapping).

   The variance $\text{Var}(D_i \mid r_1, W_i = 1)$ is also informative: if limited variation in r_2 only one type produces each r_1, then $\text{Var}(D_i \mid r_1, W_i = 1) \approx h(1-h)$ (Bernoulli variance). Excess variance beyond $h(1-h)$ indicates mixture heterogeneity in h at that rate.

3.6 Summary: Identification Landscape

Object	**Status**	**What's missing**
α, λ	Not point-identified	Require clean type recovery (Step 2), which fails without r_2
$g_{r_2\|W=2}(\cdot)$	Point-identified	Directly from bank-2 winning rates
$g(r_2)$ (population)	Not point-identified	Need $\Pr(W = 2\|r_2)$, which requires (α, λ, F)
$E[D\|r_2, W=2]$	Point-identified	Identified as a function of r_2
$F(h\|\mathbf{x})$	Not point-identified	Requires (α, λ) and clean type recovery
$\Pr(W=2)$	Point-identified	Sample proportion

The upshot: without parametric restrictions, we identify the marginals and conditional moments of the observable distribution, but cannot disentangle the latent variables $(h_i, r_2^*(\mathbf{x}_i))$ for bank-1 winners. This motivates the parametric approach in the next section.

4 Parametric Identification

We now show that imposing parametric structure on two specific objects—the type distribution and the mixing distribution—restores identification, and we characterize the conditions under which this holds.

4.1 Parametric Specification

Assumption 7 (Parametric specification). (i) **Type distribution.** $h \mid \mathbf{x} \sim F(h;\, \theta_F(\mathbf{x}))$, where F belongs to a known parametric family and $\theta_F(\mathbf{x})$ varies with $\mathbf{x}$ through a finite-dimensional parameterization.

Example: $h \mid x \sim \text{Beta}(a,\, b(x))$ on $[h_{\min}, h_{\max}]$, with $b(x) = b_0 + b_1 x$, giving $\theta_F = (a, b_0, b_1, h_{\min}, h_{\max})$.

(ii) **Covariate distribution.** $\mathbf{x} \sim P(\mathbf{x};\, \psi)$ belongs to a known parametric family.

Example: $x \sim \text{LogNormal}(\mu_x, \sigma_x^2)$, giving $\psi = (\mu_x, \sigma_x)$.

When the econometrician observes some covariates $\mathbf{z} \subseteq \mathbf{x}$, the conditioning extends: $\theta_F(\mathbf{x}, \mathbf{z})$ and $P(\mathbf{x}|\mathbf{z}; \psi)$.

The full parameter vector is $\theta = (\alpha, \lambda, \theta_F, \psi)$.

Remark 8 (Reduced-form objects induced by the primitives). The primitives $(F(h|\mathbf{x}; \theta_F),\ P(\mathbf{x}; \psi))$ determine the equilibrium objects that enter the likelihood:

- **Entrant rate.** For each $\mathbf{x}$, bank 2's equilibrium rate $r_2^*(\mathbf{x})$ is determined by the FOC (3), which depends on $(\alpha, \lambda, F(\cdot|\mathbf{x}))$. This defines a mapping $\mathbf{x} \mapsto r_2^*(\mathbf{x}; \theta)$.
- **Mixing distribution.** The population distribution of entrant rates $G(r_2; \theta)$ is the pushforward of $P(\mathbf{x}; \psi)$ through $r_2^*(\cdot; \theta)$. It is <u>not</u> a free object but is determined by the primitives through the equilibrium.
- **Conditional type distribution.** By the sufficient statistic result (Proposition 1), conditionally on r_2, the type distribution $F(h|r_2; \theta)$ is determined by averaging $F(h|\mathbf{x}; \theta_F)$ over all $\mathbf{x}$ that map to the same r_2.

The likelihood (Section 5) is written in terms of these derived objects $G(r_2; \theta)$ and $F(h|r_2; \theta)$, because the sufficient statistic result ensures that $\mathbf{x}$ enters the likelihood only through $r_2^*(\mathbf{x})$.

Remark 9 (Why two parametric choices are needed).

- **Assumption (i)** constrains the type distribution. Proposition 5 shows that without it, for any candidate (α, λ), there exist type distributions that rationalize the observed data. The parametric family restricts the degree of freedom in F, enabling the data to discipline (α, λ).
- **Assumption (ii)** characterizes the population distribution of $\mathbf{x}$, which determines the equilibrium distribution of competitive conditions. While r_2 is observed among bank-2 winners, those observations are <u>selected</u> (borrowers who switched), and recovering the population distribution requires knowing $\Pr(W = 2|r_2)$—which depends on all structural parameters. The parametric family anchors the covariate distribution.

Importantly, neither assumption alone suffices. The identification works through the <u>interaction</u>: the parametric $F(h|\mathbf{x})$ restricts how defaults relate to rates for each competitive environment, and the parametric $P(\mathbf{x})$ restricts which competitive environments arise and how frequently.

4.2 Restriction Counting: Why Parametric Models Are Generically Identified

Before analyzing the specific Beta specification, we establish a general result: any well-specified finite-dimensional parametric model satisfying Assumption 7 is generically identified—meaning that identification fails only on a measure-zero set in the parameter space. This follows from a restriction-counting argument combined with the structure of the data.

Available restrictions. The data provide two families of functional restrictions:

1. **From bank-2 winners:** For each r_2 in the support of bank-2 winners' rates, the model predicts the conditional default rate:

$$E[D|r_2, W=2] = \frac{\int h\,(1 - s(\sigma^*(h; r_2), r_2))\, dF(h; \theta_F(r_2))}{\int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h; \theta_F(r_2))} \tag{10}$$

   The left side is a known function of r_2 (estimable from data). The right side depends on the structural parameters $(\alpha, \lambda, \theta_F)$. If the support of r_2 has non-empty interior, this is a continuum of restrictions—infinitely many equations constraining finitely many parameters.

2. **From bank-1 winners:** The density of (r_1, D) among bank-1 winners is:

$$p(r_1, d, W=1 \mid \theta) = \int f(\tau(r_1; r_2) \mid r_2; \theta) \cdot \tau'(r_1; r_2) \cdot s(r_1, r_2) \cdot h^d (1-h)^{1-d} \cdot g(r_2; \theta)\, dr_2 \tag{11}$$

   where $f(\cdot|r_2; \theta)$ and $g(r_2; \theta)$ are the induced conditional type distribution and mixing density (Remark 8). This provides additional functional restrictions involving all parameters $(\alpha, \lambda, \theta_F, \psi)$.

3. **Selection-distorted mixing distribution:** The density of r_2 among bank-2 winners is:

$$g_{r_2|W=2}(r_2) = \frac{g(r_2; \theta) \cdot \Pr(W=2|r_2;\, \alpha, \lambda, \theta_F)}{\Pr(W=2)} \tag{12}$$

   which constrains ψ jointly with all other parameters.

4. **Aggregate switching rate:** $\Pr(W = 2) = \int\int (1 - s(\sigma^*(h; r_2), r_2))\, dF(h|r_2; \theta)\, dG(r_2; \theta)$ provides a scalar restriction on $(\alpha, \lambda, \theta_F, \psi)$.

Dimension count. The parameter vector $\theta = (\alpha, \lambda, \theta_F, \psi)$ is finite-dimensional: $\dim(\theta) = 2 + \dim(\theta_F) + \dim(\psi)$. For the Beta specification in the Monte Carlo ($\theta_F = (a, b_0, b_1)$, $\psi = (\mu_x, \sigma_x)$), $\dim(\theta) = 7$. The data provide:

- A continuum of restrictions from $E[D|r_2, W=2]$ (one per r_2 value in the support),
- A continuum from the bank-1 density $p(r_1, d, W=1)$,
- A functional restriction from $g_{r_2|W=2}(\cdot)$,

- A scalar from $\Pr(W=2)$.

The system is massively overdetermined: infinitely many independent restrictions for finitely many parameters.

Theorem 10 (Generic identification)**.** *Suppose the parametric families $F(h|\mathbf{x};\theta_F)$ and $P(\mathbf{x};\psi)$ are smooth in their parameters, and the support of r_2^* among bank-2 winners has non-empty interior. Then the set of parameter values $\theta = (\alpha, \lambda, \theta_F, \psi)$ at which the model is not identified (i.e., there exists $\theta' \neq \theta$ generating the same distribution of observables) has Lebesgue measure zero in the parameter space.*

Proof sketch. The functional restrictions above define a system of equations $\mathcal{M}(\theta) = \mathcal{D}$, where $\mathcal{M}(\theta)$ is the model-predicted functional (mapping parameters to the joint distribution of observables) and $\mathcal{D}$ is the data distribution. This is a map from $\mathbb{R}^{d_\theta}$ to an infinite-dimensional function space. By a standard transversality argument (cf. Rothenberg, 1971, Econometrica, Theorem 1), if the Jacobian $\partial\mathcal{M}/\partial\theta$ has full column rank at θ_0 (the "local identification condition"), then θ_0 is locally identified. The set where the Jacobian drops rank is defined by the vanishing of a $(d_\theta \times d_\theta)$ minor of an infinite-dimensional Jacobian, which is a polynomial condition in θ. For a smooth, non-degenerate parametric family, this polynomial is not identically zero, so its zero set has Lebesgue measure zero. □

Remark 11 (What generic identification does and does not say)*.* Generic identification means: for almost every true parameter value, MLE is consistent. It does not guarantee identification at every point—specific parameter configurations (e.g., at the boundary of the parameter space, or at isolated "observationally equivalent" points) may fail. Nor does it speak to how well-identified the model is: even if the model is identified, some parameters may be weakly identified if the Jacobian is close to singular, causing large standard errors in finite samples.

Generic identification is reassuring but imprecise. We now verify identification for the specific parametric specification used in the simulation.

4.3 Identification of the Beta Specification

We verify identification for the specification in the Monte Carlo:

$$h \mid x \sim \mathrm{Beta}(a, b(x)) \text{ on } [h_{\min}, h_{\max}], \qquad b(x) = b_0 + b_1\, x$$

with $x \in \{x_1, \ldots, x_{N_x}\}$ drawn from a discrete distribution (or, in the continuous case, from a parametric family $P(x;\psi)$). In equilibrium, each x_j maps to an entrant rate $r_2^*(x_j)$, inducing the conditional type distribution $F(h|r_2)$ and mixing distribution $G(r_2)$ as described in Remark 8. The full parameter vector is $\theta = (\alpha, \lambda, a, b_0, b_1, \psi)$, where ψ parameterizes the distribution of x.

Proposition 12 (Identification of the Beta specification)**.** *The parameter vector $\theta = (\alpha, \lambda, a, b_0, b_1, \psi)$ is identified from the joint distribution of (r_i^{win}, W_i, D_i) if the following conditions hold:*

(C1) The support of r_2 among bank-2 winners contains at least three distinct values.

(C2) $(a, b_0, b_1) \in (0,\infty) \times (0,\infty) \times \mathbb{R} \setminus \{0\}$ (the Beta shape parameter genuinely varies with x, and hence with r_2).

(C3) $\alpha > 0$ *and* $\lambda \in \mathbb{R}$.

(C4) $\Pr(W=2) > 0$ *(some borrowers switch).*

Proof. We proceed sequentially, using each data feature to pin down a subset of parameters.

Step 1: The functional restriction from $E[D|r_2, W=2]$.

Define the function $\delta(r_2) \equiv E[D|r_2, W=2]$, which is identified from the data. The model predicts:

$$\delta(r_2) = \frac{\int_{h_{\min}}^{h_{\max}} h \cdot (1 - s(\sigma^*(h; r_2), r_2)) \cdot f(h \mid r_2; \theta)\, dh}{\int_{h_{\min}}^{h_{\max}} (1 - s(\sigma^*(h; r_2), r_2)) \cdot f(h \mid r_2; \theta)\, dh} \tag{13}$$

where $f(h|r_2;\theta)$ is the induced conditional type density at entrant rate r_2 (inheriting the Beta shape from the primitive $F(h|x)$ through the equilibrium mapping), and $\sigma^*(h; r_2)$ depends on (α, λ).

This single functional equation in r_2 restricts three parameters: $(\alpha, \lambda, a, b_0, b_1)$ jointly. The key insight is that these five parameters affect $\delta(r_2)$ through distinct channels:

- (α, λ) control the <u>selection weights</u> $(1 - s)$: how much the adverse-selection reweighting distorts the mean type. Specifically, α governs the sensitivity of the switching probability to rate differences, while λ shifts the level.
- (a, b_0, b_1) control the <u>base distribution</u>: the unweighted mean $E[h|r_2]$ and its response to r_2.

By (C1), $\delta(r_2)$ is observed at three or more values of r_2. Since $\delta(r_2)$ is not linear in any of the five parameters (due to the nonlinear equilibrium map σ^* and the nonlinear Beta density), three or more values generically provide at least three independent nonlinear equations.

Moreover, the <u>shape</u> of $\delta(r_2)$ as a function of r_2 is particularly informative. By (C2), $b_1 \neq 0$, so the Beta distribution genuinely shifts with r_2. The curvature of $\delta(r_2)$ depends jointly on how the Beta distribution changes (b_1) and how the selection weights change (α). If α is large, selection is strong and δ is pulled further below $E[h|r_2]$; if α is small, $\delta \approx E[h|r_2]$ and the curvature is driven by the Beta family alone.

Step 2: The aggregate switching rate.

The data identify $\Pr(W=2)$. The model predicts:

$$\Pr(W=2) = \int\int (1 - s(\sigma^*(h; r_2), r_2)) \cdot f(h|r_2;\theta)\, dh\, dG(r_2;\theta) \tag{14}$$

This depends on $(\alpha, \lambda, \theta_F, \psi)$. Crucially, λ enters as an additive shifter in the logit: $s = 1/(1 + \exp(\alpha\Delta - \lambda))$. The aggregate switching rate is therefore monotonically decreasing in λ (holding other parameters fixed), which means that once (α, θ_F, ψ) are pinned down, λ is uniquely determined by matching $\Pr(W=2)$. This provides the primary direct restriction on λ.

Step 3: Separation of (α, λ) from θ_F.

The key separating mechanism is that α and λ affect both the <u>level</u> and the <u>within-r_2 shape</u> of the default curve $\delta(r_2)$, while θ_F affects only the <u>cross-r_2 pattern</u>. Formally:

- Fix r_2. The selection weight $(1 - s(\sigma^*(h; r_2), r_2))$ is a function of h alone (via σ^*), parameterized by (α, λ). Higher α makes this weight steeper in h (stronger selection), pulling $\delta(r_2)$ further below $E[h|r_2]$. The gap $E[h|r_2] - \delta(r_2)$ at each r_2 disciplines α.

- Across r_2 values: $\delta(r_2)$ traces a curve whose slope and curvature are governed by b_1 (how the type distribution shifts) and a (its shape). Since b_1 enters through the Beta family and α enters through the selection weights—and these are multiplicative in the integrand (13)—they generically cannot mimic each other.

Step 4: Identification of ψ.

Given $(\alpha, \lambda, \theta_F)$, the selection probability $\Pr(W = 2|r_2)$ and the equilibrium mapping $x \mapsto r_2^*(x)$ are both determined. The induced population density of r_2 must satisfy:

$$g_{r_2|W=2}(r_2) = \frac{g(r_2; \theta) \cdot \Pr(W = 2|r_2)}{\Pr(W = 2)}$$

Since $g_{r_2|W=2}$ is known from data and $\Pr(W = 2|r_2)$ is now determined, $g(r_2; \theta) \propto g_{r_2|W=2}(r_2)/\Pr(W = 2|r_2)$. This pins down the population distribution of r_2, and since $r_2^*(x)$ is monotone, it pins down the distribution of x and hence ψ.

Step 5: Overidentification from bank-1 winners.

Steps 1–4 use only bank-2 winner moments and the aggregate switching rate. The entire distribution of (r_1, D) among bank-1 winners, given by (11), must also be matched by the same θ. This provides a rich set of overidentifying restrictions. In particular, the bank-1 data independently constrain (α, λ) through the shape of the winning-rate distribution and the conditional default rates $E[D|r_1, W = 1]$. Any discrepancy between the parameter values implied by bank-1 vs. bank-2 moments signals misspecification. □

Remark 13 (The role of $b_1 \neq 0$). Condition (C2) requires that the type distribution genuinely varies with r_2. If $b_1 = 0$, the Beta distribution is the same at every r_2, and the function $\delta(r_2)$ varies only through the selection weights (α, λ). In this case, $\delta(r_2)$ traces the composition of a fixed Beta distribution under varying selection, which restricts $(\alpha, \lambda, a, b_0)$ but reduces the number of independent restrictions. Identification may still hold (from bank-1 moments), but it is less robust.

Remark 14 (Necessity of three r_2 values). Condition (C1) requires at least three distinct r_2 values among bank-2 winners. With only one r_2 value, we are in the M5 setting and identification follows from the nonparametric argument (no parametric F is needed). With two r_2 values, $\delta(r_2)$ provides two restrictions, which combined with $\Pr(W = 2)$ gives three equations for five parameters $(\alpha, \lambda, a, b_0, b_1)$—generically insufficient. Three values provide the additional curvature information needed to separate the selection effect (α) from the distribution shift (b_1).

Remark 15 (Practical verification via Monte Carlo). The propositions above are population identification results: they state that the true parameter is the unique maximizer of the population log-likelihood. Whether MLE can recover the parameters in finite samples depends on the strength of identification (the curvature of the likelihood near the optimum). The Monte Carlo exercise in Section 6 provides the practical verification: if MLE consistently recovers the true parameters across replications and parameter configurations, this confirms both identification and adequate finite-sample performance.

4.4 Intuition: What Each Data Feature Identifies

Parameter(s)	Primary identifying variation
ψ (distribution of $\mathbf{x}$)	Distribution of winning rates among bank-2 winners, after correcting for selection. Since $r_2^*(\mathbf{x})$ maps covariates to entrant rates, the population distribution of r_2 (and hence ψ) is recovered by inverting the selection distortion: $g(r_2;\theta) \propto g_{r_2\|W=2}(r_2)/\Pr(W=2\|r_2)$. The selection correction depends on $(\alpha, \lambda, \theta_F)$, so ψ is jointly identified with all other parameters.
θ_F (type distribution)	**How default rates vary with r_2 among bank-2 winners.** If the default rate rises steeply as r_2 increases (riskier borrowers face higher entrant rates), this reveals how the type distribution shifts with r_2. The shape of this curve disciplines θ_F through equation (10).
α (price sensitivity)	**The severity of adverse selection.** α controls how much the selection function $1 - s(h)$ varies with h. High α: strong selection, bank-2 winners are much better types than bank-1 winners. Low α: weak selection, default rates are similar across winners. The gap $E[D\|W=2, r_2] - E[D\|W=1, r_1]$ across matched rate levels traces out α.
λ (switching cost)	**The overall retention rate.** λ shifts bank-1's share uniformly: higher λ means $\Pr(W = 1)$ is higher at every rate differential. The aggregate bank-1 share $\Pr(W=1)$ provides a direct moment, and conditional shares $\Pr(W=1\|r_2)$ provide a functional restriction.

4.5 Conditions for Identification Failure

Even under Assumption 7, identification can fail in degenerate cases:

1. **No bank-2 winners.** If λ is very large, $\Pr(W = 2) \approx 0$ and the data contain almost no bank-2 winners. Without bank-2 winners, we lose the direct observation of r_2^* and the conditional default rates that discipline θ_F and ψ. In the limit $\Pr(W=2) = 0$, only bank-1 rates and defaults are observed, and the model collapses to a single-agent pricing problem where (α, λ) enter only through a single index $\lambda - \alpha\Delta$—a classic underidentification problem.

2. **No variation in r_2^*.** If $\mathbf{x}$ has no effect on r_2^* (or $\mathbf{x}$ is degenerate), the model reduces to M5 where bank 2 plays a single rate. In this case, identification follows from the M5 argument and the parametric assumptions are unnecessary—but if we incorrectly estimate a heterogeneous model, ψ compresses to a point mass and θ_F effectively loses its r_2-dependence, which may cause numerical difficulties.

3. **Overparameterization of θ_F.** If the type distribution $F(h|\mathbf{x};\theta_F)$ has too many free parameters relative to the identifying variation (e.g., allowing flexible higher moments that the data cannot discipline), the model may be locally unidentified even if globally consistent.

4.6 Overidentifying Restrictions

Under the parametric specification, the model is typically overidentified: the data provide more restrictions than there are free parameters. Key testable restrictions:

1. **Consistency of bank-1 and bank-2 moments.** The parameters $(\alpha, \lambda, \theta_F)$ are disciplined by bank-2 winners (through (10)) and by bank-1 winners (through (11)). Any discrepancy between the parameter values implied by each subsample indicates model misspecification.

2. **Adverse selection direction.** The model predicts $E[D|W = 2, r_2] > E[h|r_2]$ (bank-2 winners are riskier-than-average types, because $1 - s$ is increasing in h). This is testable: if bank-2 winners have lower default rates than bank-1 winners with similar rates, the adverse selection story is contradicted.

3. **Linearity of the identifying restriction within strata.** When conditioning on r_2 (observed for bank-2 winners), the M5 identifying restriction $\ln(\alpha m - 1) = \lambda - \alpha\Delta$ should hold. Bank-2 winners reveal r_2, and if we could observe the bank-1 rate that each bank-2 winner would have received, we could check this directly. Instead, the restriction is embedded in the likelihood through $\sigma^*(h; r_2)$.

4. **Bank-2's FOC.** Having estimated $(\alpha, \lambda, \theta_F)$, the model predicts a specific $r_2^*(r_2)$ at each covariate value—i.e., the bank-2 FOC (3) should be satisfied. The predicted distribution of r_2^* must match the observed distribution among bank-2 winners (after selection correction).

5 Likelihood

The likelihood follows from the identification argument.

5.1 Bank-1 Winners ($W_i = 1$)

The entrant rate r_2 is latent; we marginalize over G:

$$\boxed{\mathcal{L}_i^{(1)}(\theta) = \int_{\mathcal{R}_2(r_i^{\text{win}})} f\big(\tau(r_i^{\text{win}}; r_2) \mid r_2; \theta\big) \cdot \tau'(r_i^{\text{win}}; r_2) \cdot s(r_i^{\text{win}}, r_2) \cdot h^{D_i}(1-h)^{1-D_i} \cdot g(r_2; \theta)\, dr_2} \tag{15}$$

with $h = \tau(r_i^{\text{win}}; r_2)$.

5.2 Bank-2 Winners ($W_i = 2$)

The entrant rate is observed ($r_i^{\text{win}} = r_2$); the type h is latent:

$$\boxed{\mathcal{L}_i^{(2)}(\theta) = g(r_i^{\text{win}}; \theta) \cdot \int_{\underline{h}}^{\bar{h}} \big(1 - s(\sigma^*(h; r_i^{\text{win}}), r_i^{\text{win}})\big) \cdot h^{D_i}(1-h)^{1-D_i} \cdot f(h \mid r_i^{\text{win}}; \theta)\, dh} \tag{16}$$

5.3 Full Log-Likelihood

$$\ell(\theta) = \sum_{i=1}^{N} \Big[\mathbf{1}(W_i = 1) \ln \mathcal{L}_i^{(1)}(\theta) + \mathbf{1}(W_i = 2) \ln \mathcal{L}_i^{(2)}(\theta)\Big] \tag{17}$$

5.4 Computational Strategy

1. Pre-compute $\sigma^*(h; r_2)$ on a fine grid of (h, r_2) values.
2. Construct $\tau(r_1; r_2)$ by numerical inversion on the same grid.
3. Evaluate the integrals in (15)–(16) by quadrature.
4. Key efficiency: the equilibrium grid depends only on (α, λ), not on (θ_F, ψ). When only θ_F or ψ changes, reuse the pre-computed grid.

6 Monte Carlo Simulation

The identification theory is validated through Monte Carlo simulation. The code is in `Code/model6/`.

6.1 Data Generating Process

Grids. $x \in \{x_1, \ldots, x_{N_x}\}$ and $h \in \{h_1, \ldots, h_{N_h}\}$ on equally spaced grids.

Joint distribution. $P(x = x_j, h = h_k)$ specified as an $N_x \times N_h$ probability matrix. The conditional $P(h|x)$ is constructed using a Beta density with shape parameter depending on x, so that higher x implies riskier borrowers. Parameters are set in `params.m`.

Equilibrium. Solved for each x_j using the nested algorithm (iterate on r_2^* with damped best-response updates; within each iteration, solve bank-1's FOC by Brent's method for each h_k).

Data generation. N borrowers drawn from $P(x, h)$; equilibrium rates assigned; winner drawn from logit shares; default drawn from Bernoulli(h).

6.2 Identification Checks

1. **Bank-2 rate constancy within x.** Among bank-2 winners with the same x, all winning rates should be identical ($r_i^{\text{win}} = r_2^*(x)$). Confirms equilibrium is correctly computed.

2. **Type recovery (conditional on x).** Among bank-1 winners, $E[D_i \mid r_i^{\text{win}}, x_i]$ should equal h_i (the M5 argument applies within each x-stratum).

3. **Linear identifying restriction.** $\ln(\alpha m - 1) = \lambda - \alpha\Delta$ should hold across all (h, x) pairs. Verified on population and simulated data.

4. **Adverse selection.** $E[D|W=2, x] > E[D|W=1, x]$ for each x value.

6.3 Estimation Exercise

The identification analysis is tested by:

1. Simulating data under multiple parameter configurations (varying α, λ, θ_F).

2. Estimating the model via MLE for each configuration.

3. Comparing estimated to true parameters.

The driver script `run_identification.m` automates this, saving simulation and estimation results per configuration. If the model is identified, the MLE should recover the true parameters as N grows.

7 Discussion

7.1 Comparison of Identification Across Models

	M5 (no heterogeneity)	**M6 with x** (covariates observed)	**M6 without x** (this document)
r_2 observation	Single constant	Constant within $\mathbf{x}$	Distribution
Type recovery	Exact: $E[D\|r_1, W=1] = h$	Exact within $\mathbf{x}$: $E[D\|r_1, W = 1, \mathbf{x}] = h$	Mixture: $E[D\|r_1, W=1]$ is a weighted avg
(α, λ)	Nonparametric (linear restriction)	Nonparametric (overidentified)	Parametric (via F, G families)
F	Nonparametric	Nonparametric within $\mathbf{x}$	Parametric $F(h\|\mathbf{x}; \theta_F)$
Key assumption	Logit demand	Logit demand	Logit + parametric F, G

7.2 Sensitivity to Parametric Assumptions

The parametric assumptions in Assumption 7 are the price of not observing $\mathbf{x}$. Several strategies can test sensitivity:

1. **Flexible families.** Use distributions with enough parameters to nest multiple shapes (e.g., 4-parameter beta, or a mixture of betas for F; a mixture of normals for G). If estimates are stable across flexible specifications, the parametric assumption is not driving the results.

2. **Partial observability.** When some covariates $\mathbf{z} \subseteq \mathbf{x}$ are observed, conditioning on $\mathbf{z}$ tightens the mixtures and reduces dependence on parametric assumptions. Comparing estimates with and without conditioning on $\mathbf{z}$ reveals how much the parametric assumptions matter.

3. **Specification tests.** The overidentifying restrictions in Section 4.6 provide formal tests of the parametric model. Rejection indicates the parametric family is too restrictive.

7.3 Extensions

1. **Multi-product data.** If the same borrower has multiple loan products, the common $(h, \mathbf{x})$ generates correlated rates across products, providing additional identification.

2. **Panel data.** Within-borrower variation in r_2^* (driven by $\mathbf{x}$ changes over time) can be used to condition on r_2, reducing the mixture problem.

3. **Semiparametric approaches.** Rather than fully parametric F and G, one could use sieve estimation or Bayesian nonparametric priors (e.g., Dirichlet process mixtures) to flexibly approximate the distributions while maintaining enough structure for identification.